Exchangeability and de Finetti's theorem

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May 27, 2026

Chapter 1

Introduction

This blueprint documents the formalization of **de Finetti's theorem** and the **de Finetti–Ryll-Nardzewski equivalence** for infinite sequences on *standard Borel spaces*.

The main result establishes a three-way equivalence between:

- **Contractable**: All strictly increasing subsequences of equal length have the same distribution
- **Exchangeable**: Distribution invariant under finite permutations
- **Conditionally i.i.d.**: There exists a probability kernel such that finite marginals equal mixtures of product measures

We formalize *all three proofs* from Kallenberg (2005):

1. **Koopman/Ergodic approach** using the Mean Ergodic Theorem
2. **$\mathbf{L}^2$ approach** using elementary contractability bounds
3. **Martingale approach** using reverse martingale convergence (after Aldous)

Chapter 2

Foundations

2.1 Core Definitions

Definition 1 (Exchangeable sequence)**.** A sequence $(X_n)_{n\in\mathbb{N}}$ of random variables is *exchangeable* with respect to measure μ if for every $n \in \mathbb{N}$ and every permutation σ of $\{0, \dots, n-1\}$, the joint distribution of $(X_{\sigma(0)}, \dots, X_{\sigma(n-1)})$ equals that of $(X_0, \dots, X_{n-1})$.

Definition 2 (Contractable sequence)**.** A sequence $(X_n)_{n\in\mathbb{N}}$ is *contractable* with respect to measure μ if for all $m \in \mathbb{N}$ and all strictly increasing functions $k, k' : \mathrm{Fin}(m) \to \mathbb{N}$, the joint distribution of $(X_{k(0)}, \dots, X_{k(m-1)})$ equals that of $(X_{k'(0)}, \dots, X_{k'(m-1)})$.

Definition 3 (Conditionally i.i.d. sequence)**.** A sequence $(X_n)_{n\in\mathbb{N}}$ is *conditionally i.i.d.* with respect to measure μ if there exists a probability kernel $\nu : \Omega \to \mathrm{Measure}(\alpha)$ such that for any strictly monotone $k : \mathrm{Fin}(m) \to \mathbb{N}$, the joint distribution of $(X_{k(0)}, \dots, X_{k(m-1)})$ equals the mixture $\mu.\mathrm{bind}(\omega \mapsto \nu(\omega)^{\otimes m})$.

2.2 σ-algebra Infrastructure

Definition 4 (Tail σ-algebra)**.** The *tail σ-algebra* of a sequence (X_n) is $\mathcal{T} = \bigcap_{n=0}^{\infty} \sigma(X_n, X_{n+1}, \dots)$.

Definition 5 (Future filtration)**.** The *future filtration* at level m is $\mathcal{F}_m = \sigma(X_{m+1}, X_{m+2}, \dots)$.

Lemma 6 (Future filtration is antitone)**.** *The future filtration is antitone: $m \le n$ implies $\mathcal{F}_n \le \mathcal{F}_m$.*

Lemma 7 (Tail σ-algebra contained in future filtration)**.** *For all m: $\mathcal{T} \le \mathcal{F}_m$.*

Definition 8 (Shift-invariant σ-algebra)**.** The (one-sided) shift-invariant σ-algebra on path space $\mathbb{N} \to \alpha$ consists of measurable sets S such that $\theta^{-1}(S) = S$, where θ is the left shift. A two-sided counterpart on $\mathbb{Z} \to \alpha$, `ViaKoopman.shiftInvariantSigmaℤ`, is introduced separately as scaffolding for the Koopman route's lag-constancy lemmas.

Chapter 3

Easy Directions

3.1 Exchangeable implies Contractable

Lemma 9 (Permutation extension)**.** *Any strictly increasing function* $k : \mathrm{Fin}(m) \to \mathbb{N}$ *with range contained in* $\{0, \ldots, n-1\}$ *extends to a permutation of* $\{0, \ldots, n-1\}$*.*

Theorem 10 (Exchangeable implies Contractable)**.** *If* (X_n) *is exchangeable, then it is contractable.*

3.2 Conditionally i.i.d. implies Exchangeable

Theorem 11 (Conditionally i.i.d. implies Exchangeable)**.** *If* (X_n) *is conditionally i.i.d., then it is exchangeable.*

Chapter 4

Main Implication: Contractable implies Conditionally i.i.d.

This is the deep direction of de Finetti's theorem. We formalize three independent proofs.

4.1 Via Martingale (Aldous' proof)

The martingale approach uses reverse martingale convergence to the tail σ-algebra.

4.1.1 Pair Law Equality

Lemma 12 (Pair law equality for contractable sequences). *For a contractable sequence and $k \leq m$, the joint distribution of $(X_k, X_{m+1}, X_{m+2}, \ldots)$ equals that of $(X_m, X_{m+1}, X_{m+2}, \ldots)$.*

Lemma 13 (Contractable distribution equality). *For a contractable sequence, the joint distribution of $(X_k, \theta_{m+1}X)$ equals that of $(X_m, \theta_{m+1}X)$ for all $k \leq m$, where θ_n is the shift operator.*

Lemma 14 (Conditional expectation of indicator equals under contractability). *For contractable sequences and $k \leq m$: $\mathbb{E}[\mathbf{1}_{X_k \in B} \mid \mathcal{F}_m] = \mathbb{E}[\mathbf{1}_{X_m \in B} \mid \mathcal{F}_m]$ a.s.*

4.1.2 Kallenberg Chain and Convergence

Lemma 15 (Kallenberg chain lemma). *Conditional expectations of indicators given the reverse filtration converge to conditional expectations given the tail σ-algebra.*

Lemma 16 (Conditional expectation convergence). *For $k \leq m$ and measurable B: $\mathbb{E}[\mathbf{1}_{X_m \in B} \mid \mathcal{F}_m] = \mathbb{E}[\mathbf{1}_{X_k \in B} \mid \mathcal{F}_m]$ a.s.*

Lemma 17 (Extreme members equal on tail). *For any measurable B: $\mathbb{E}[\mathbf{1}_{X_m \in B} \mid \mathcal{T}] = \mathbb{E}[\mathbf{1}_{X_0 \in B} \mid \mathcal{T}]$ a.s.*

4.1.3 Factorization and Directing Measure

Lemma 18 (Finite level factorization)**.** *For finite products of indicators, the conditional expectation given $\mathcal{F}_m$ factors as a product of individual conditional expectations.*

Lemma 19 (Tail factorization from future)**.** *The tail σ-algebra factorization follows from the finite-level factorization via reverse martingale convergence.*

Lemma 20 (Block coordinate conditional independence)**.** *Coordinates are conditionally independent given the tail σ-algebra.*

Lemma 21 (Directing measure is probability measure)**.** *The directing measure $\nu(\omega)(B) = \mathbb{E}[\mathbf{1}_{X_0 \in B} \mid \mathcal{T}](\omega)$ is a probability measure for every ω (the bound is pointwise, not merely a.e., via mathlib's* `isProbability_condExpKernel`*).*

Lemma 22 (Directing measure measurability)**.** *The directing measure $\omega \mapsto \nu(\omega)(B)$ is measurable for each Borel B.*

Lemma 23 (Conditional law of X_n equals directing measure)**.** *Per-coordinate identification: for every n and every Borel B, $(\omega \mapsto \nu(\omega)(B).\mathrm{toReal}) =^{\mu\text{-}a.e.} \mathbb{E}[\mathbf{1}_B \circ X_n \mid \mathcal{T}]$. This is the bridge condition consumed by 24 below.*

Lemma 24 (Finite product formula)**.** *Under contractability, for every strictly monotone selection $k : \mathrm{Fin}\, m \to \mathbb{N}$ and any kernel ν satisfying the per-coordinate bridge 23:*

$$\mathrm{Measure.map}\,(\omega \mapsto (i \mapsto X_{k\,i}(\omega)))\,\mu \;=\; \mu.\mathrm{bind}\,(\omega \mapsto \mathrm{Measure.pi}\,(\lambda\, i,\; \nu(\omega))).$$

This is the martingale route's analogue of the L^2/Koopman 49: it bundles the rectangle π-system extension and the Measure.map = bind *identification into one step driven by per-coordinate conditional expectations.*

Theorem 25 (Contractable implies Conditionally i.i.d. (via Martingale))**.** *If (X_n) is contractable, then it is conditionally i.i.d. The directing kernel $\nu(\omega)(B) = \mathbb{E}[\mathbf{1}_{X_0 \in B} \mid \mathcal{T}](\omega)$ is constructed from the tail σ-algebra, identified per-coordinate by 23, and lifted to the joint law by 24 — bypassing the L^2/Koopman common-ending step.*

4.2 Via L^2 (Elementary proof)

The L^2 approach uses elementary contractability bounds on block averages. This is Kallenberg's "second proof" and has the lightest dependencies.

Note: This proof applies to *real-valued* sequences ($X : \mathbb{N} \to \Omega \to \mathbb{R}$) with L^2 integrability (i.e., $\mathbb{E}[X_i^2] < \infty$ for all i).

4.2.1 Block Averages and Covariance Structure

Definition 26 (Block average (L^2 version))**.** The block average $A_n = \frac{1}{n}\sum_{i=0}^{n-1} f(X_i)$ for bounded measurable f.

Lemma 27 (Contractable covariance structure)**.** *For contractable sequences, the covariance $\mathrm{Cov}(f(X_i), f(X_j))$ is constant for $i \neq j$.*

Lemma 28 (L^2 contractability bound)**.** *For contractable sequences, certain L^2 norms of block averages are bounded.*

4.2.2 Cesaro Convergence

Recorded Kallenberg L^2 bound. Kallenberg's Lemma 1.2 gives an L^2 distance bound for weighted averages of an *exchangeable* sequence. It is recorded in the Lean library as `Exchangeability.DeFinetti.ViaL2.` for completeness, but it is not on the active proof path — the route below uses the contractability-only variant 28 so that the proof of Contractable ⇒ Conditionally i.i.d. does not assume exchangeability.

Lemma 29 (Block averages are Cauchy in L^2). *For contractable X and bounded f, the block Cesaro averages* $\mathrm{blockAvg}\, f\, X\, 0\, n$ *form a Cauchy sequence in L^2.*

Lemma 30 (Cesaro to conditional expectation (L^2)). *Block averages converge in L^2 to the conditional expectation given the tail.*

Lemma 31 (Cesaro to conditional expectation (L^1)). *Block averages converge in L^1 to the conditional expectation given the tail.*

4.2.3 Directing Measure Construction

The L^2 route now goes through mathlib's `condExpKernel` machinery: the Stieltjes construction supplies the candidate directing measure $\omega \mapsto \nu(\omega)$, and a pair of bridge lemmas identifies its push-forward and integral with the kernel coming from mathlib's `condExpKernel` on the tail σ-algebra. The shared common-ending theorem then lifts the indicator-level identity to the full product σ-algebra.

Lemma 32 (Directing measure is probability (L^2 version)). *The Stieltjes measure constructed from the limiting CDF is a probability measure for every ω (pointwise, not merely a.e.). Monotonicity, bounds, and right-continuity of the underlying CDF are inherited directly from mathlib's `stieltjesOfMeasurableRat` construction.*

Theorem 33 (Directing measure equals `condExpKernel` push-forward). *Almost everywhere in ω, the Stieltjes-built directing measure coincides with the push-forward of mathlib's `condExpKernel` $\mu\, \mathcal{T}\, \omega$ by X_0. This is the kernel-level bridge from the analytic L^2 construction to mathlib's conditional-expectation kernel API.*

Lemma 34 (Directing-measure integral equals conditional expectation). *For a bounded measurable function f, the integral $\int f\, d\nu(\omega)$ against the directing measure equals $\mathbb{E}[f \circ X_0 \mid \mathcal{T}](\omega)$ almost everywhere. This is the shape consumed by the shared common-ending step.*

Theorem 35 (Weighted sums converge in L^1). *The weighted sums of indicators converge in L^1.*

Theorem 36 (Contractable implies Conditionally i.i.d. (via L^2)). *If (X_n) is contractable (and real-valued), then it is conditionally i.i.d.* ***Scope note:*** *the L^2 route is currently stated for real-valued sequences via the Stieltjes-based directing-measure construction; extending it to general standard Borel targets would require a measure-disintegration replacement for the Stieltjes step.*

4.3 Via Koopman (Mean Ergodic Theorem)

The Koopman approach uses the Mean Ergodic Theorem via the shift operator on L^2. This is Kallenberg's "first proof" and uses disjoint-block averaging.

Scope note: the Koopman route is currently stated for real-valued sequences (the Mean Ergodic Theorem is invoked through $\mathrm{L}^2(\mu; \mathbb{R})$ on path space). Generalising to standard Borel targets would require an L^2-valued ergodic infrastructure beyond what is presently formalised. The default project export (50) handles general standard Borel spaces via the martingale route.

4.3.1 Block Averages and Ergodic Theory

Definition 37 (Block average)**.** The *block average* $A_{m,n,k}(f)(\omega) = \frac{1}{n}\sum_{j=0}^{n-1} f(\omega_{k\cdot n+j})$ averages f over the k-th block of size n (indices $[kn, kn+n)$). For $n = 0$, the block average is defined as 0.

Lemma 38 (Koopman-condexp commutation)**.** *The conditional expectation operator onto the shift-invariant subspace commutes with the Koopman operator.*

Theorem 39 (Birkhoff averages converge to condexp)**.** *Birkhoff averages converge in L^2 to the conditional expectation given the shift-invariant σ-algebra.*

Lemma 40 (Block averages converge in L^1)**.** *For a shift-invariant measure, block averages converge in L^1 to the conditional expectation given the shift-invariant σ-algebra: $\int |A_{m,n,k}(f) - \mathbb{E}[f \circ \pi_0 \mid \mathcal{I}]|\, d\mu \to 0$ as $n \to \infty$.*

4.3.2 Contractability and Factorization

Lemma 41 (Conditional expectation lag constancy from exchangeability)**.** *For exchangeable sequences, the conditional expectation of a product does not depend on the lag between coordinates.*

Lemma 42 (Integral product equals block average product)**.** *For contractable sequences, integrals of products factor through block averages.*

Lemma 43 (Product block average L^1 convergence)**.** *Products of block averages converge in L^1.*

Theorem 44 (Conditional expectation product factorization)**.** *For contractable sequences, the conditional expectation of a product of indicators factors as a product of conditional expectations.*

Lemma 45 (Bridge from contractability)**.** *For contractable sequences, indicator products satisfy the bridge condition.*

Theorem 46 (Contractable implies Conditionally i.i.d. (via Koopman))**.** *If (X_n) is contractable, then it is conditionally i.i.d. This proof uses the Mean Ergodic Theorem via the Koopman operator on L^2.*

Chapter 5

Common Ending

The L^2 and Koopman routes share a common final step: a directing kernel built from the tail (or shift-invariant) σ-algebra is fed to a π-system / monotone-class extension argument that lifts the indicator-level identity to the full product σ-algebra. The martingale route reaches conditional independence through its own parallel `finite_product_formula` pipeline (built on `condExpKernel`) rather than the lemmas in this chapter.

5.1 Comparing the two endings

The L^2/Koopman ending (49, in `CommonEnding.lean`) and the martingale ending (24, in `ViaMartingale/FiniteProduct` are similar in total length and both finish with a rectangle π-system extension via `MeasureTheory.ext_of_generate_fin` The substantive difference is the *shape of the bridge condition* each accepts about the directing kernel ν.

Route	Bridge shape	How the route produces it
Martingale	Per-coordinate conditional-expectation equality: $(\omega \mapsto \nu(\omega)(B).\mathrm{toReal}) =^{\text{a.e.}} \mathbb{E}[\mathbf{1}_B \circ X_n \mid \mathcal{T}]$ for every n, B (see 23).	Falls out directly from mathlib's `condExpKernel` applied to the tail σ-algebra.
L^2/Koopman	Joint integral equality on the whole m-tuple: for every injective $k : \mathrm{Fin}\, m \to \mathbb{N}$ and rectangle B, $\int^{-} \prod_i \mathbf{1}_{X_{k\,i} \in B_i}\, d\mu = \int^{-} \prod_i \nu(\omega)(B_i)\, d\mu$.	Falls out of Cesàro / ergodic averaging — both sides are limits of the same averaged sequence.

Why this matters. Each route's directing-measure construction naturally produces the bridge shape its own ending expects, so the routes pick the ending that matches what they already have. The martingale route's per-coordinate bridge is "smaller" (one -algebra-relative equality per index) and is what `condExpKernel` delivers without extra work; the common ending's joint integral bridge is what limit-of-averages arguments deliver without extra work, but would be wasteful if you already had per-coordinate CE equalities. At the stitching level both routes compress to a short top-level theorem that hands a 4-tuple $(\nu, h_{\nu\text{-prob}}, h_{\nu\text{-meas}}, h_{\text{bridge}})$ to the chosen ending — the choice of ending is what differs, not the amount of stitching.

Lemma 47 (π-system uniqueness). *Measures on product spaces are determined by their finite-dimensional marginals.*

Theorem 48 (Monotone class theorem). *The monotone class theorem allows extending from indicators to measurable functions.*

Theorem 49 (Conditional independence extension). *Conditional independence on indicators extends to the full product σ-algebra.*

Chapter 6

Main Theorem

We package the three-way equivalence as three separate top-level theorems, one per proof of the main implication. The default project export is the martingale version.

Theorem 50 (de Finetti–Ryll-Nardzewski equivalence (martingale route))**.** *For an infinite sequence* $(X_n)_{n\in\mathbb{N}}$ *of random variables taking values in a* standard Borel space α *(with* α *nonempty), the following are equivalent:*

1. (X_n) *is contractable*
2. (X_n) *is exchangeable*
3. (X_n) *is conditionally i.i.d. (i.e., there exists a directing kernel* ν*)*

This is the default project export. The proof constructs ν *from the tail* σ*-algebra* $\mathcal{T}$ *via* $\nu(\omega)(B) = \mathbb{E}[\mathbf{1}_{X_0\in B} \mid \mathcal{T}](\omega)$.

Theorem 51 (de Finetti–Ryll-Nardzewski equivalence (L^2 route))**.** *Same three-way equivalence, proved via the elementary* L^2 *approach. This route is currently stated for real-valued sequences (see the* L^2 *section caveat).*

Theorem 52 (de Finetti–Ryll-Nardzewski equivalence (Koopman route))**.** *Same three-way equivalence, proved via the Mean Ergodic Theorem through the Koopman operator. This route is currently stated for real-valued sequences (see the Koopman section caveat).*